See First, Refuse When Thin: Compiling Point-in-Time World Bundles for LLM Agents in Crypto Markets

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Abstract

Purpose. In AI-based finance, the performance frontier is often set less by model complexity than by whether the market world fed to the model is broad, stable, and honest. We treat the missing piece as a *data end*: an anonymized observation runtime that compiles multi-band crypto evidence into a point-in-time (PIT) *world bundle* (BandPIT/WMI) with hard refusal when support is thin—infrastructure for LLM/agent market analysis and for evaluating whether models *understand* that world, not a trading engine. **Evidence.** Primary—seeing: grounding recovers ready/missing/tilt at ≈0.97 under Compiled vs. ≲0.23 under Raw, while sufficiency chat alone is unreliable (Compiled mean 0.60). Secondary—valve: under a *frozen full-schema* stress gate (max WMI ≈ 0.093<0.2), Compiled thin-abstain is 1.0 (100-day and full OOS n=2000) while Ungated soft judgment is only ≈0.68 / ≈0.56; Raw over-trades. Secondary—handoff: the *scoped-archive production valve* opens on ≈61% OOS days; live LLMs abstain 1.0 on closed days and act on ≈89% of open days, while a no-LLM tilt rule remains nonempty (Sharpe 0.77). **Claim.** First, give models a world; then ask them to decide. We move from model-first to *LLM-consumable* data-end-first: compile so models can read the market state; refuse when they must not act—a machine-native contract, not a human dashboard.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics.*

Keywords

AI in finance, LLM agents, data end, observation layer, world bundle, grounding, market understanding, cryptocurrency, point-in-time, quality-gated refusal

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1 Introduction

Purpose (read this first). At the methodology–application interface of AI and finance, many LLM/agent failures are not "the model is not smart enough," but "the market worldview is too thin." The stack is **data/observation → decision → execution** (Table 1); agents and generative world models [6, 7, 9, 19–21] typically assume the first stage. Crypto violates that assumption: exchange, macro, and alternative evidence arrive on incompatible clocks [11, 13].

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Table 1: AI-finance stack; this paper supplies the data end for LLM/agent consumers.

Stage	Typical owner	This paper
Data / observation	Assumed by agents	World bundle + quality valve
Decision	LLM / policy / portfolio	Out of scope
Execution	Broker / OMS	Out of scope

Stress episodes with multi-hour exchange outages make the gap concrete: an agent that sees only lagged momentum can act while the observation set is incomplete. We build the missing substrate—compile a PIT world bundle so models can *see* the market, evaluate that seeing with checkable grounding, and hard-refuse when the view is too thin. The product is reduced world-model error and auditable AI judgments, not trading answers.

Failure modes. No feed → refuse (safe, useless as a data end). Thin price fragment → trade and lose (unsafe for any downstream agent). Disclosed but ungated world → read fields, disagree on action. These are not model failures; they are interface failures. Primary consumer: the LLM agent—not the human desk—so the bundle replaces visual intuition with typed readiness/age/quality fields and a hard abstain boolean. So the data end must (i) make the market machine-legible to LLMs/agents and (ii) enforce a quality valve before decision/execution.

What we build. An anonymized *data-provision runtime for public LLMs and finance agents* (Section 4): multi-band evidence → BandPIT/WMI → JSON world bundle for GPT /DeepSeek /GLM /Gemini-class consumers, served through a read API. It is also an evaluation substrate: the same bundle supports grounding and gated-action protocols that test whether a model understands disclosed market state. "World model" means state compiler + quality gate (Section 3), not dynamics and not a decision engine. The bundle is complete, honest, and auditable; the hard abstain flag is the machine-readable valve.

How we evaluate (dual protocol). **RQ1 (primary—seeing):** Are ready /missing /tilt answers verifiable against PIT ground truth under Compiled vs. Raw? **RQ2 (valve):** Under a *frozen full-schema stress gate* (never clears 0.2), does hard policy enforce thin abstention where soft MAY does not? Arms: Compiled, HPO (hard prompt on numeric WMI, no boolean), Ungated, Raw, Blind. **RQ3a (handoff):** Under the *scoped-archive production valve*, do live LLMs obey closed days and act when open? Does a transparent no-LLM rule remain callable? **RQ3b (content):** On the same vintaged tilts, does an ungated content probe beat momentum (ΔCE; LOBO)?

Contributions. (1) AI–finance data-end semantics: epistemic observations, compilation Π_t, world-quality index with band scope, and seeing vs. gating—interfaces that let LLM/agent stacks consume crypto markets safely. (2) An anonymized data-provision path for agents: collectors to BandPIT to service surface (bundle / health / handoff) to world bundles, plus a frozen four-arm harness to *evaluate* market understanding. (3) Live LLM evidence that the substrate works: grounding ≈0.97; stress-gate abstain 1.0; scoped production valve obedience (closed 1.0; open act ≈0.89); nonempty tilt handoff (Sharpe 0.77; LOBO 1.0). Estimand: whether public models can see a typed market world, refuse when thin, and hand off when ready—not portfolio alpha.

2 Related Work

World models. World models learn compact states and often dynamics [6, 7, 9]. We address the complementary observation-side problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy—not a Dreamer/JEPA simulator.

LLM agents in finance (ICAIF intersection). Return prediction [12], open financial LLMs [18], memory- and tool-augmented agents [10, 19–21], and ICAIF retrieval/debate/judge systems [1, 3, 15, 16] improve how agents fetch, debate, and are scored. Methodologically they still assume a usable observation stream; application stacks then conflate bad market context with bad policy. We target that gap in a data-centric sense: compile a typed PIT world for LLM/agent consumers, and score market understanding (grounding, evidence-bound refusal) before debating alpha. The estimand is orthogonal to memory routers and debate judges: without a typed world, those systems evaluate agents on tool fragments, not on whether the market view was broad, stable, and honest. We return to this contrast in Section 6.

Selective prediction and PIT. Abstention can be optimal under noise [4, 5]; we attach refusal to *world* quality $q(W_t)$ via a runtime field, not classifier confidence alone (Prop. 3.10)—blocking the overconfident-when-thin failure mode. Macro vintages [2] and crypto microstructure [11, 13] motivate multi-band compilation with disclosed missingness (Prop. 3.8). Bootstrap discipline for the content probe follows [8, 14, 17].

3 Observation-Layer Semantics

This section states the data-end theory as *interface semantics*: what counts as an observation, how compilation yields a world, and when quality forces refusal. It is not a pricing theorem and not a generative dynamics model. Systems realization is Section 4; empirical tests are Section 5.

Definition 3.1 (Epistemic observation). An observation is the tuple $O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t})$ (value, latest-available time, quality, main-view gate, semantic role). Omitting (τ, q, g, r) yields a feature dump, not a world observation.

Definition 3.2 (Compilation). $\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views into a consumer object.

Definition 3.3 (World bundle). $W_t = \Pi_t(\mathcal{F}_t^{\mathrm{raw}})$ with scalar quality $q(W_t)$; $\mathcal{F}_t^{\mathrm{AI}} = \sigma(W_t)$. W_t is the handoff object to decision.

Definition 3.4 (World-quality index). Ready, limited, and missing counts disclose coverage. We write $q(W_t) \equiv \mathrm{WMI}_t$ for brevity in theoretical statements. $\mathrm{WMI}_t = B_t U_t H_t$ aggregates breadth, stability, and honesty in $[0, 1]$; ACWMI multiplies a regime weight. Band scope selects the support of (B, U, H): full-schema stress audit vs. the declared consumer archive (production valve). Scoped WMI scores only declared bands; it is a contract-completeness index, not a lowered $q^\star$. The Compiled interface exposes a hard abstain boolean $\mathbf{1}\{q(W_t) < q^\star\}$ (and, under archive scope, incompleteness) with a thin-world label (production $q^\star = 0.2$ for scoped WMI). We further define the Evidence Alignment Ratio (EAR) as the fraction of rationale evidence cues that bind to disclosed bundle observations (IDs/bands/tilts), measuring how tightly decisions attach to the world.

Seeing vs. gating. The bundle is what the model *sees*; the hard flag *blocks* action when $q(W_t) < q^\star$. Tilts without a gate are a dump; a gate without a readable bundle is only a switch. The two interfaces are separable: Ungated keeps W_t and numeric $q(W_t)$ but removes the hard boolean. This is not feature engineering: a feature vector asks the model to infer reliability; a world bundle *states* clocks, missingness, and a machine-readable valve.

Assumption 3.5 (Bounded lag and increments). Band j has max observation lag Δ_j; latent increments satisfy $\|S_u - S_{u-1}\| \le \bar{\delta}$.

Assumption 3.6 (Multiplicative quality independence). World availability requires breadth, stability, and honesty *jointly*: if any factor is 0, then $\mathrm{WMI}_t = 0$. The product $\mathrm{WMI} = BUH$ is the simplest continuous $[0, 1]$ score obeying this axiom; averages can stay large when one factor vanishes.

Remark 3.7. *Practically, we treat* WMI $< \varepsilon$ *(e.g.,* ε=10^{-3}*) as effective zero, consistent with the binary abstain contract.*

Proposition 3.8 (Lag–missingness reconstruction bound). *Under Assumption 3.5, any compiler that uses only in-clock observations satisfies*

$$\|\widetilde{S}_t - S_t\| \;\le\; C_{\mathrm{del}}\bar{\delta}\sum_j \Delta_j + C_{\mathrm{noi}}\varepsilon_t + C_{\mathrm{miss}}m_t,$$

where m_t is disclosed missing mass and ε_t collects measurement noise. Operationally, BandPIT maps age to ready/limited/missing so m_t and lag terms are readable in W_t; compilation cannot erase delay, but it can refuse to pretend the bound is zero. With C_{del}=1 under worst-case band alignment, typical crypto parameters (Δ_{ex}≈2d, Δ_{macro}≈5d, $\bar{\delta}$≈ daily volatility ≈3%) make the lag term alone large relative to daily moves—illustratively $O(30\%)$ of daily-return variance scale—justifying refusal when B, U, H are incomplete rather than treating $q^\star$=0.2 as a pure empirical knob.

Proposition 3.9 (Compilation ≠ dump). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\mathrm{AI}}$: ungated stale evidence can raise volume while lowering honesty, tightening rather than loosening Prop. 3.8.*

Proposition 3.10 (World-conditional abstain). *Let $c_{\mathrm{act}}(W_t)$ be the expected cost of acting and c_{abs} the abstain cost. If $c_{\mathrm{act}}(W_t) > c_{\mathrm{abs}}$ and c_{act} is nonincreasing in $q(W_t)$, the Bayes action is abstain on $\{W : q(W) < q^\star\}$ [4]. A hard runtime contract implements that set; soft "you may abstain" text does not. In finance, this aligns with the* *classic "know when not to trade" principle; our contribution is binding it to observable world quality rather than model confidence.*

Remark 3.11 (Threshold $q^\star = 0.2$). *Production valve: archive complete and* WMI $\geq q^\star$*. For* $q^\star \in [0.10, 0.30]$ *open share stays* $\approx$0.61 *(thick days* ≈ 1*; gap days fail completeness; gap median* $\approx$0.14*). A WMI-only counterfactual moves when* $q^\star$ *crosses that gap mass. We keep* 0.2 *above gap-day WMI—also where Ungated soft thin-abstain fails (*$\approx$0.68*; Sec. 5.3).*

Proposition 3.12 (LOBO content vs. gating). *Deleting band* j *changes a downstream value functional through a content channel and a gating channel. Under an ungated rule the gating channel is shut, so the value gap identifies nonempty fields.*

Theory $\rightarrow$ *protocol.* Prop. 3.8–3.9 $\Rightarrow$ RQ1. Prop. 3.10 $\Rightarrow$ RQ2 (Compiled/HPO hard policy vs. Ungated soft MAY). Def. 3.4+Rem. 3.11 $\Rightarrow$ RQ3a (scoped valve). Prop. 3.12 $\Rightarrow$ RQ3b.

Remark 3.13 (Not a generative WM). *We do not learn* $p(s_{t+1} \mid s_t, a_t)$*. We deliver a state compiler with a quality-gated refuse interface: the data end of the stack.*

4 Runtime: Data-End Compile Path

The anonymized prototype implements the data end as a layered compile path for LLM/agent consumers (Figure 1, Table 2). Decision and execution sit downstream; this paper supplies their market context and the protocols that test whether that context was understood. Paths and product names are omitted for double-blind review.

4.1 Layered design

(1) **Data layer.** Exchange, macro, and alternative collectors write vintage-aware history (observation timestamps; macro `available_at`).
(2) **Store layer.** Embedded stores hold raw series, merged market panels, and compiled analytics (readiness, WMI/ACWMI, world snapshots).
(3) **Logic layer.** BandPIT reconstructs readiness on a previous-close clock; honesty and missingness gates implement Π_t; WMI, ACWMI, and availability shocks O_t are first-class.
(4) **Service layer.** A read API serves quality-tagged bundles and health objects without restarting collectors for newly arrived rows. Anonymized surface includes world-bundle reads, a valve-gated handoff endpoint (tilt action, no LLM), health exposing the abstain flag with band scope, a paper world-model time slice, and availability-shock queries.
(5) **Consumer /eval layer.** Frozen-prompt public LLMs (and rule agents) call an OpenAI-compatible adapter at temperature 0 under four arms—the harness that scores market understanding against the supplied world.

4.2 Evaluation archive

The runtime schema is band-extensible; this study evaluates the three-band consumer object in Table 3 on a PIT panel of $\approx$3990 asset-days (2025-07 to 2026-08): exchange ready share $\approx$0.80 panel-wide ($\approx$0.61 OOS), macro 1.0, alternative 1.0. **Full-schema stress WMI** scores permanently empty non-archive slots and caps at max $\approx$0.093, so a 0.2 gate never opens—the frozen LLM-arm protocol (RQ2). **Scoped-archive production WMI** renormalizes B, U, H to {exchange, macro, alternative} and opens iff the archive is complete and WMI $\geq$ 0.2 (RQ3a); this measures contract readiness, not a relaxed threshold.

Table 2: Compile-path modules (systems view, not a product inventory).

Module	Role
Collectors / stores	Timestamped / vintaged band history
BandPIT compiler	Previous-close readiness; Π_t
Quality indices	$B, U, H \rightarrow$ WMI; band scope; abstain; O_t
Bundle builder	Complete / honest / auditable JSON
Service surface	Bundle / health / handoff / availability reads
Consumer harness	Four-arm protocol; temp. 0

Table 3: Input evidence vs. fields emitted in the world bundle (live archive).

Band	Fetched (raw)	Emitted (bundle)
exchange	OHLCV, funding, OI	mom5; status/age
macro	equity-vol / dollar indices	macro_tilt; status/age
alternative	stablecoin circulation	alt_tilt; status/age
derived	—	WMI, ACWMI, abstain, O_t, evidence IDs

4.3 PIT clock and freshness

For calendar day t, BandPIT reconstructs the decision information set at $(t-1)$ 23:59 (payoff is same-day close-to-close return); the bundle records `decision_asof`. Each band is graded ready /limited /missing against daily-scale age thresholds (Table 4). Missingness is disclosed rather than imputed (Prop. 3.8). Breadth, stability, and honesty aggregate into $\text{WMI}_t = B_t U_t H_t$ (ACWMI adds regime weight). PIT-safe tilts use only in-clock observations: five-day macro changes, a seven-day stablecoin-flow slope, and mom5. The *scoped production valve* sets `should_ai_abstain` when the archive is incomplete or scoped WMI < 0.2, and exposes `thin_world`; bundles also disclose `full_schema_wmi` as a ceiling audit. Figure 2 (top): macro and alternative stay ready; exchange readiness varies and defines thick/thin days. Figure 2 (bottom): same panel, two scopes—full-schema stress WMI stays under 0.2; scoped production WMI clears 0.2 on archive-complete days (shaded). Table 5: scoped production valve opens on $\approx$61% OOS days ($\equiv$ 3/3 ready); full-schema stress gate stays closed.

4.4 World bundle (what the LLM sees)

The consumer object is a JSON world bundle (Table 6): timing, completeness, honesty, quality (including the abstain flag), content tilts, and `evidence_ids`. Table 7 contrasts human-desk intuition with the typed fields an LLM must parse. Listing 1: exchange-gap day (scoped valve closed). Listing 2: 3/3-ready day where scoped WMI=1.0 opens the production valve; `full_schema_wmi`$\approx$0.093 is disclosed as ceiling audit—same day, two scopes, not a lowered $q^\star$.

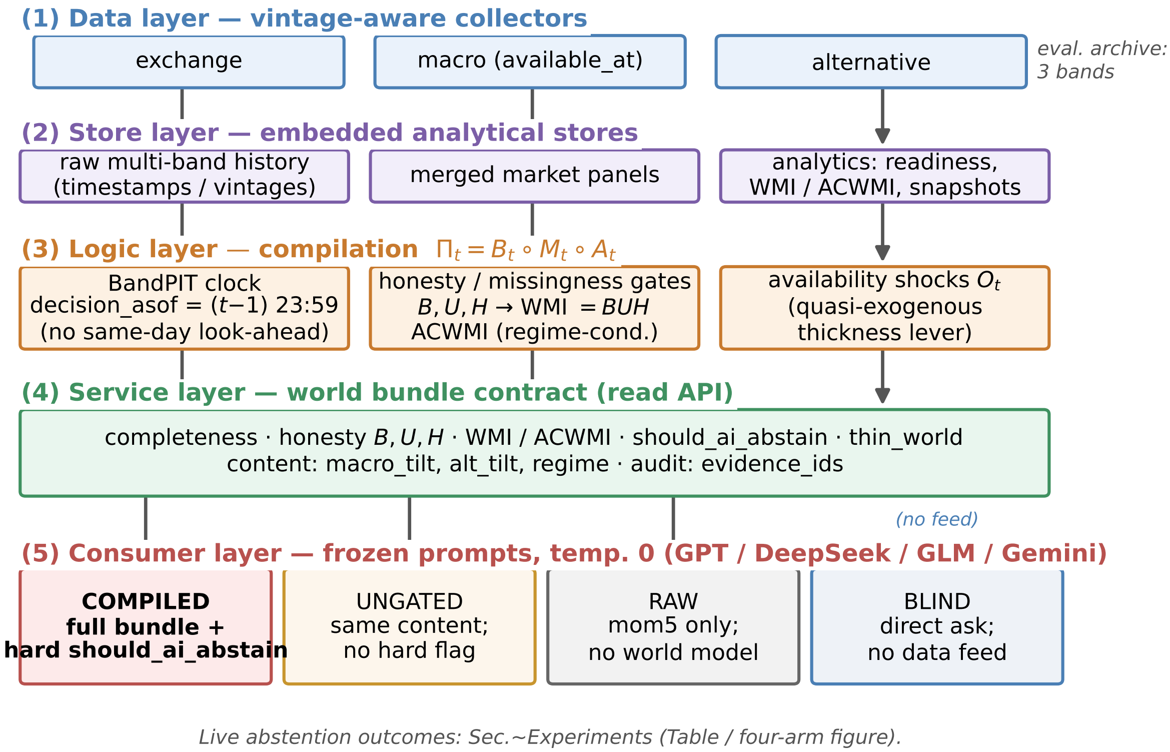

Figure 1: Data-end pipeline: evidence → BandPIT → world bundle → Compiled / Ungated / Raw / Blind.

Table 4: BandPIT freshness thresholds (seconds → days).

Band	Fresh window	In eval archive
exchange	2d	yes (ready ≈0.80)
macro	5d	yes (ready 1.0)
alternative	7d	yes (ready 1.0)
news / on-chain / options	3d	missing
tokenomics	7d	missing
event calendar	14d	missing

Table 5: Dual gates vs. open share (OOS). Production = scoped; stress = full schema.

Gate	Open share	Note
Scoped production 0.2	0.61	complete ∧ WMI≥ 0.2
Full-schema stress 0.2	0.00	max WMI ≈0.093

Table 6: World-bundle field groups (the market view).

Group	Fields
Timing	decision_asof, previous-close protocol
Completeness	ready/limited/missing counts, band status
Honesty	B, U, H; main-view / stale gates
Quality	WMI, ACWMI, band scope, abstain, thin-world
Content	macro_tilt, alt_tilt, mom5
Audit	evidence IDs; EAR; full-schema WMI

Table 7: Human desk vs. LLM agent consumption of the same market day.

Human desk	LLM agent
Reads candlesticks	Reads band_status
Feels a "thin" world	Reads WMI=0.14
Decides to wait	Reads should_ai_abstain: true
Scans news/context	Matches evidence_ids

The bundle omits charts/colors; EAR and dual-scope WMI are symbolic contracts LLMs process natively.

Listing 1: Exchange-gap bundle (Compiled; scoped closed).


```
{"date":"2026-05-11","asset":"ADA",
 "decision_asof":"2026-05-10T23:59:00",
 "completeness":{"n_ready":2,"n_missing":1},
 "world_model_index":{"band_scope":"eval_archive",
   "wmi":0.139,"archive_complete":false,
   "should_ai_abstain":true,"thin_world":true,
   "full_schema_wmi":0.015},
 "macro_tilt":1.0,"alt_tilt":-1.0}
```


Listing 2: 3/3-ready day: scoped valve opens.


```
{"date":"2026-01-16","asset":"ADA",
 "completeness":{"n_ready":3,"n_missing":0},
 "world_model_index":{"band_scope":"eval_archive",
   "wmi":1.0,"archive_complete":true,
   "should_ai_abstain":false,"thin_world":false,
   "full_schema_wmi":0.093},
 "band_status":{"exchange":"ready","macro":"ready",
   "alternative":"ready"},
```

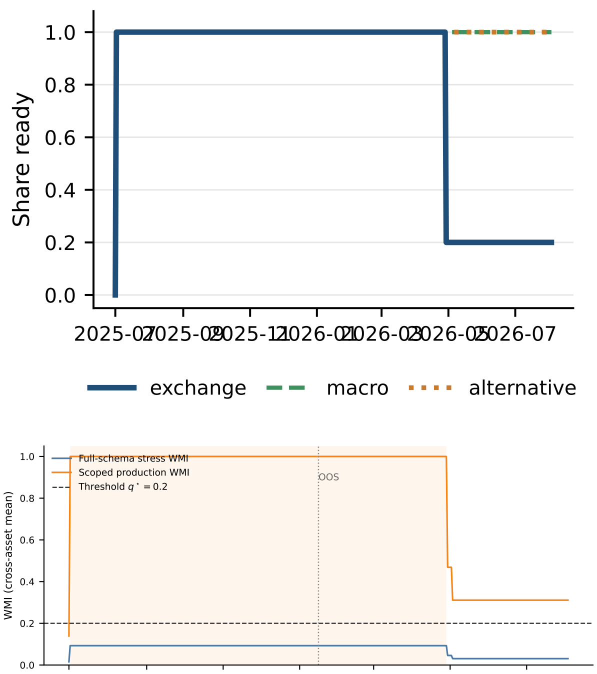

Figure 2: Top: PIT readiness in the archive. Bottom: dual-scope WMI (stress ceiling vs. production contract; shaded = archive-complete).

```
"macro_tilt":1.0,"alt_tilt":-1.0}
```

4.5 Arms and frozen prompts

Actions ∈ {bullish, bearish, neutral, abstain}. **Compiled:** full bundle; prompt *requires* abstain when the hard boolean is true. **HPO** (hard-prompt-only): same numeric fields as Ungated (*no* boolean); prompt *requires* abstain if WMI < 0.2. **Ungated:** same content and numeric WMI; soft prompt (*may* abstain). **Raw:** mom5 only. **Blind:** date+asset, no feed. **Dual protocol.** RQ1–RQ2 freeze full-schema bundles (max WMI ≈ 0.093); RQ3a switches to scoped production bundles. Prompts, IS/OOS cut, and seeds are frozen (Listings 3–4).

Listing 3: Compiled (excerpt): hard boolean.

```
1. If should_ai_abstain is true OR the world is thin,
   you MUST "abstain".
2. Prefer band content over raw momentum.
```

Listing 4: Ungated (excerpt): soft judgment.

```
1. Judge whether low WMI makes action unsafe;
   you MAY "abstain".
NO hard boolean; NO forced policy.
```

Why the hard valve is not a confound. The data end must *block* thin-day action (Prop. 3.10). HPO separates hard *prompt* obedience on numeric WMI from the typed boolean; Ungated removes both; Raw/Blind bound the ladder.

4.6 Systems validation (engines)

Paper-lab smoke wires BandPIT/WMI/ACWMI and checks the dual gate: scoped archive opens on 3/3-ready statuses while full-schema stays closed; a handoff rule acts only when the scoped valve is open. Availability shocks O_t are queryable on the service surface.

5 Experiments

5.1 Setup

PIT panel ~399 days × 10 liquid names (≈3990 asset-days; 2025-07–2026-08), chronological IS/OOS 200/200 days (cut 2026-01-16). Unless noted, live arms share the same stratified 100 OOS asset-days (seed fixed), temperature 0, and four flash/mini vendors (gpt-5.4-mini, deepseek-v4-flash, glm-5.2, gemini-3.5-flash-lite) via an OpenAI-compatible adapter. Grounding uses 50 OOS days; full-OOS abstention uses n=2000; no-LLM probes use ≈200 OOS days after return alignment. Primary metrics: RQ1 grounding F1/tilt accuracy (sufficiency chat secondary); RQ2 thin-world abstain under the *full-schema stress gate* (CE as avoided loss, not alpha); RQ3a scoped production-valve LLM obedience plus no-LLM tilt handoff; RQ3b ungated tilt ΔCE vs. momentum.

5.2 RQ1 (primary): Seeing—grounding workflow

If the data end works, models must report a *checkable* market view—not invent coverage from a price fragment.

On 50 OOS asset-days, each model lists ready/missing bands among exchange/macro/alternative, reports macro_tilt / alt_tilt signs, and judges sufficiency; (i)–(iii) score against PIT ground truth.

Table 8: Compiled mean ready/missing F1 and tilt-sign accuracy 0.97; Raw collapses to 0.04, 0.23, 0.19. Raw has no readiness or tilt fields, so it cannot recover archive state; Compiled forces use of the disclosed world.

Legibility ≠ sufficiency chat. Table 9: Compiled sufficiency accuracy averages only 0.60 (Gemini 0.04) despite 0.97 field grounding, while Raw often answers "enough?" correctly by chance/heuristics (0.95) while inventing coverage. Chat judgment of "is this enough to trade?" is not a substitute for a typed world; we therefore measure seeing by checkable fields (RQ1) and action safety by the hard valve (RQ2).

Case study. Listings 5–6: ADA (2026-05-11) exchange missing; Compiled recovers the checkable view; Raw cannot. A second day (2026-01-17) under the *full-schema stress* Compiled arm abstains on `should_ai_abstain`; Raw goes bullish from mom5 = +1 alone.

Listing 5: Compiled grounding (gpt-5.4-mini, ADA 2026-05-11).

```
ready: alternative, macro
missing: exchange
macro_tilt +; alt_tilt -; sufficiency: insufficient
"thin world... should_ai_abstain=true"
```

Listing 6: Raw grounding / action (same model family).

```
grounding: ready=[]; cannot confirm tilts
action (2026-01-17): bullish
"Positive 5-period momentum in raw feed."
```

Table 8: Grounding (50 days): primary seeing result. C = Compiled; R = Raw.

Model	Ready F1 C	Ready F1 R	Missing F1 C	Missing F1 R	Tilt-sign C	Tilt-sign R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Table 9: Sufficiency chat vs. field grounding (50 days). Seeing = fields; chat alone misleads.

Model	Suff. C	Suff. R	Ready F1 C	Ready F1 R
gpt-5.4-mini	0.78	0.98	0.94	0.00
deepseek-v4-flash	0.78	0.90	0.98	0.06
glm-5.2	0.80	0.96	0.98	0.09
gemini-3.5-flash-lite	0.04	0.94	0.96	0.03
Mean	0.60	0.95	0.97	0.04

Table 10: Thin abstention on identical 100 OOS days under full-schema stress. Blind = 1.0; HPO = hard prompt on numeric WMI, no boolean.

Model	C	HPO	U	Abs. R	ΔCE
gpt-5.4-mini	1.00	1.00	0.63	0.04	+0.73
deepseek-v4-flash	1.00	1.00	0.81	0.75	+1.22
glm-5.2	1.00	1.00	0.86	0.11	+1.04
gemini-3.5-flash-lite	1.00	1.00	0.43	0.07	+1.30
Mean	1.00	1.00	0.68	—	+1.07

5.3 RQ2 (safety valve): Hard refuse when the view is thin

Legibility does not imply safe action. All live LLM arms here freeze *full-schema* bundles, so every OOS day is thin under the stress gate 0.2. Table 10 and Figure 3: on the same 100 days, Compiled and HPO abstain 1.0; Ungated soft-judges at mean 0.68 (Gemini 0.43); Raw over-trades (Table 11); Blind abstains 1.0 with no feed—safe but useless as a data end. HPO recovers perfect thin abstention when full-schema WMI is unambiguously below 0.2; Ungated soft MAY does not. Under production-scoped WMI (Sec. 5.4), where values span $[0.09, 1.0]$, HPO closed compliance softens to ≈0.96 (vendor range 0.93–1.0) while the boolean valve stays 1.0; thus the typed boolean is essential for cross-threshold contract stability, not merely stress scenarios. Offline soft evidence-alignment (rationale cues to disclosed bands/tilts/quality) averages ≈0.998 Compiled / 0.993 HPO vs. 0.80 Ungated and 0.38 Raw on matched 100-day slices. Mean ΔCE (Compiled−Raw) +1.07 is avoided loss, not alpha.

Full-OOS replication. Table 12: on all 2000 OOS asset-days (all thin under the *full-schema stress* gate 0.2), Compiled thin-abstain remains 1.000 (Wilson [.998, 1]) for every vendor. Ungated soft rates fall to a four-vendor mean ≈0.56; Raw GPT/Gemini abstain at 0.000.

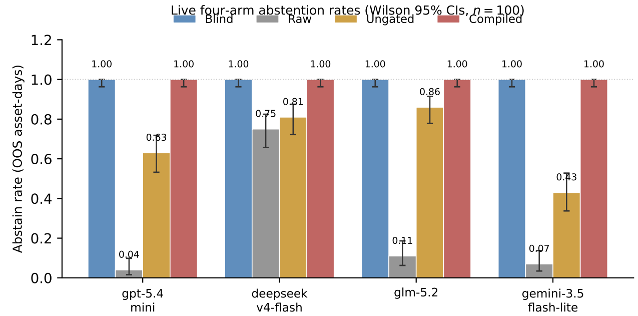

Figure 3: Safety valve: Blind refuses; Raw over-trades; Ungated is unstable; Compiled/HPO enforce hard policy.

Table 11: Raw action mix (100 days): half-baked context drives directional mass.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Table 12: Full-OOS thin abstention (n=2000; all thin under full-schema stress gate 0.2).

Model	Thin C	Thin U	Abs. R
gpt-5.4-mini	1.000	0.666	0.000
deepseek-v4-flash	1.000	0.540	0.748
glm-5.2	1.000	0.679*	0.078*
gemini-3.5-flash-lite	1.000	0.359	0.000
Mean	1.000	≈0.56	—

*GLM coverage incomplete (Ungated n=1402; Raw n=675).

Scale does not rescue soft judgment. Blind refuses everywhere with no feed on the paired 100-day sample (abstain 1.0 for every vendor): the failure mode that needs a data end is not "models never abstain," but "thin fragments make them act."

Inside Ungated soft acts. On the mixed 100-day sample, when Ungated *does* act, act-conditional CE is sharply negative for three of four vendors (DeepSeek −3.12, GLM −4.28, Gemini −1.67) while GPT is positive on its smaller act set. Soft disclosure without a hard flag is not a free lunch: the same readable world that RQ1 scores can still harm decisions when the valve is removed.

Open slice (still under full-schema LLM arms). Table 13: on the band-thick subset (3/3 ready ≡ full-schema WMI ≥ 0.05; ≈61% OOS; n≈1224) Ungated keeps compiled content without the hard boolean—mean abs. ≈0.48, CE positive for every vendor (mean ≈+0.29; bootstrap ≈+0.31, CIs include 0). Full-schema Compiled still abstains at 1.0 here; Raw stays risky (GPT/Gemini abs. 0.0, CE negative). The same thick days are where the *scoped production valve* opens; live scoped LLM handoff is RQ3a.

Table 13: Open slice under full-schema LLM arms (3/3 ready; $n\approx1224$). Soft-judgment lower bound.

Model	Abs. U	CE U	Act-CE U	Abs. R	CE R
gpt-5.4-mini	0.58	+0.42	+0.32	0.00	−0.28
deepseek-v4-flash	0.42	+0.21	+0.45	0.75	+0.04
glm-5.2	0.58	+0.24	+0.11	0.09*	−0.87*
gemini-3.5-flash-lite	0.34	+0.31	+0.06	0.00	−0.25

*Raw GLM n=398; Ungated GLM n=932; others n=1224. Bootstrap mean CE ≈+0.31 (B=5, 999 reps; one-sided $p\approx0.21$). Slice-specific Abs. U (mean ≈0.48); compares to mixed 100-day mean 0.68 in Table 10.

Table 14: RQ3a scoped production valve (70 open + 30 closed): Compiled boolean vs. HPO.

Model	Abs C_{cl}	Abs H_{cl}	Act C_{op}	Act H_{op}	ΔCE_{op}	ΔCE_{cl}
gpt-5.4-mini	1.00	0.97	0.97	1.00	−4.09	+0.49
deepseek-v4-flash	1.00	1.00	0.76	0.66	−5.02	+0.17
glm-5.2	1.00	0.93	0.89	0.94	−3.92	+3.06
gemini-3.5-flash-lite	1.00	0.93	0.93	0.86	−3.82	+2.64
Mean	1.00	0.96	0.89	0.86	−4.21	+1.59

Abs/Act: closed abstain / open act. H=HPO (no boolean). $\Delta CE_{op/cl}$ = Compiled−Raw (open usability; closed avoided loss).

Table 15: RQ3a no-LLM handoff. Scoped production valve vs. full-schema stress.

Policy (reads bundle fields)	Sharpe	CE†	Abstain
Full-schema stress 0.2	0.000	0.000	1.00
Scoped production 0.2 + tilts	**0.772**	**+0.253**	0.39
Momentum (mom5)	0.101	−0.075	0.00
Buy-and-hold	−1.399	−0.987	0.00

†CE = ann. mean − $\frac{1}{2}$ ann. variance (valve-aware path).

5.4 RQ3a (handoff): Scoped production valve with live LLMs

Having established seeing (RQ1) and stress-gate refusal (RQ2), we ask whether the *production* valve is callable. The full-schema stress gate never clears 0.2, so it cannot open—by design in RQ2. Scoped production WMI opens on ≈61% OOS days (≡ 3/3 ready). We re-run the same four vendors on a stratified sample of 70 open + 30 closed OOS asset-days with scoped bundles (seed fixed; temp. 0).

Table 14: on closed days Compiled abstains 1.0 for every vendor while HPO (numeric WMI + hard prompt; no boolean) softens to mean ≈0.96; on open days Compiled acts ≈89% vs. HPO ≈86%. Open-day ΔCE vs. Raw is negative (mean ≈−4.2): a usability check that models act when the valve opens, not a trading metric (no-LLM tilt Sharpe 0.772, Table 15). Closed-day refusal avoids Raw losses (mean ΔCE≈+1.6).

A transparent no-LLM rule on the same vintaged tilts remains the clean nonempty-field handoff (Table 15): Sharpe 0.772 / CE +0.253 when the scoped valve is applied, versus momentum and buy-and-hold; full-schema stress stays fully closed.

5.5 RQ3b (content): Nonempty fields

Table 16 and Figure 4: ungated always-on tilts vs. momentum on the same vintaged fields (ΔCE=0.334, p=0.034; LOBO content share 1.0). This probe shuts the gating channel (Prop. 3.12); it is not the production valve.

Table 16: RQ3b ungated content probe (no valve). Bootstrap ΔCE 0.334.

Policy	Sharpe	CE‡
Buy-and-hold	−1.404	−1.163
Momentum	0.096	−0.206
World-bundle tilts	**0.764**	**+0.130**
Tilts − Momentum	—	0.334 (p=0.034)

‡Content-harness CE (always-on; not comparable 1:1 to †).

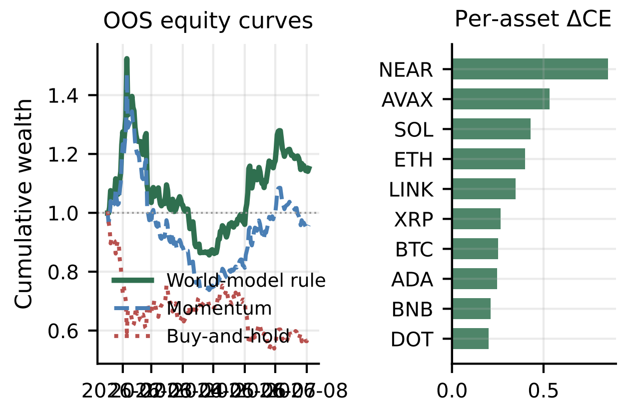

Figure 4: Secondary content probe: compiled tilts beat momentum; ΔCE positive for every asset.

Table 17: RQ3b content probe by archive density (OOS; ungated).

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.334*
Open / thick	0.61	0.061	0.315
Thin (exchange gap)	0.39	1.048§	0.129

*Block-bootstrap ΔCE; raw mean gap 0.336. §Always-on probe trades thin days by construction; high CE is a selection effect, not a reason to remove the valve.

Density split and LOBO channel. Table 17: open/thick ΔCE=0.315; on exchange-gap days the always-on probe still trades (high slice CE is a selection artifact, not a valve claim; ΔCE=0.129). Table 18 and Figure 5 (Prop. 3.12): deleting macro or alternative collapses the gap through the content channel (share 1.0; gating residual 0).

6 Discussion

What carries the paper. (i) Machine-native legibility for an LLM-consumable data end: grounding ≈0.97 from a typed, auditable object—not a human-simplified view (sufficiency chat is not a substitute). (ii) Hard policy binds on thin days (Compiled/HPO 1.0); soft MAY does not (≈0.68). (iii) Scoped production opens ≈61% OOS days; LLMs obey the valve; open-day CE ≠ alpha; no-LLM tilts stay nonempty. Relative to FinMem /FinAgent and ICAIF agent

Table 18: LOBO decomposition (ungated content probe). Content share 1.0.

Drop	ΔCE content	p	Content share
macro	−0.334	0.034	1.0
alternative	−0.334	0.034	1.0
exchange	0.000	1.000	—

Exchange drop leaves macro/alt tilts unchanged, so content share is undefined (—); zero gap ≠ "exchange is uninformative."

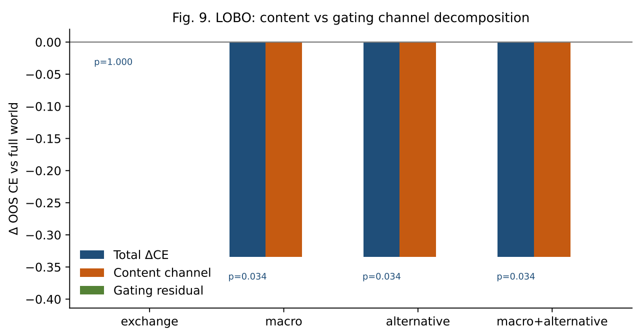

Figure 5: LOBO: deleting macro/alternative removes the edge via content, not the refuse flag.

stacks [1, 3, 15, 16, 20, 21], we supply the observation stage those systems assume.

Deployment path. Level 1: observation API (PIT world bundles; this paper). Level 1 is already implementable today: any LLM agent can replace its raw OHLCV feed with a BandPIT read-API call and gain automatic thin-day refusal without changing decision logic. Level 2: community benchmark for market understanding (grounding, gated action, soft EAR). Level 3: compliance logging of abstain decisions for algorithmic accountability. We do not claim to solve crypto prediction or open-day LLM alpha; seeing a typed world reduces decision risk before models get "smarter."

Design and scope. World bundle ≠ feature dump; previous-close PIT; dual WMI scope; HPO separates hard prompt from hard boolean. Feature engineering outputs vectors; we output typed worlds with clocks, missingness, and a machine-readable valve. Prompts, cut, and seeds are frozen; an anonymized harness ships on acceptance.

7 Conclusion

AI-in-finance LLM/agent stacks need a reliable data end before decision or execution. We formalize that observation layer and show: grounding ≈0.97; hard policy (Compiled/HPO) thin-abstains 1.0 while soft Ungated does not; scoped production opens and is obeyed; tilts remain nonempty without claiming open-day LLM alpha. We do not teach models to be smarter. We teach them what the market looks like in a language they can parse—and when not to guess. In AI finance, seeing the world correctly is a prerequisite, not a luxury. Move from model-first to LLM-consumable data-end-first: compile so models can read; refuse when they must not act.

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